E. Boneca Ambalabu

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1200 | Tags: bitmasks | https://codeforces.com/problemset/problem/2094/E

Boneca Ambalabu gives you a sequence of n integers $a_1,a_2,\ldots,a_n$.

Output the maximum value of $(a_k \oplus a_1)+(a_k \oplus a_2)+\ldots+(a_k \oplus a_n)$ among all $1 \le k \le n$. Note that $\oplus$ denotes the bitwise XOR operation.

Input

The first line contains an integer t ($1 \le t \le 10^4$) – the number of independent test cases.

The first line of each test case contains an integer n ($1 \le n \le 2 \cdot 10^5$) – the length of the array.

The second line of each test case contains n integers $a_1,a_2,\ldots,a_n$ ($0 \le a_i$ lt 2^{30}).

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, output the maximum value on a new line.

Example

Input

```
5
3
18 18 18
5
1 2 4 8 16
5
8 13 4 5 15
6
625 676 729 784 841 900
1
1
```

Output

```
0
79
37
1555
0
```

Note

In the first test case, the best we can do is $(18 \oplus 18)+(18 \oplus 18)+(18 \oplus 18)=0$.

In the second test case, we choose $k=5$ to get $(16 \oplus 1)+(16 \oplus 2)+(16 \oplus 4)+(16 \oplus 8)+(16 \oplus 16)=79$.